



1. Setup & Motivation

Score-based black-box attacks see probabilities, not gradients. Standard untargeted losses mainly lower the true-class confidence:

$$\mathcal{L}_{\text{untarg}}(x', y) = P(y \mid x') \quad \text{or} \quad \log P(y \mid x').$$

- Probability mass is freed, but no non-true class is chosen.
- The trajectory can drift across several class basins.
- Query budget is wasted before misclassification.

Goal. Pick a viable target from the trajectory, then commit early to avoid drift.

Gap.

- Targeted attacks need a target before the run starts.
- CE/probability losses often do not track a competitor.

2. Method: Opportunistic Target Selection

OTS keeps the proposal mechanism unchanged. Only the loss schedule changes after a short untargeted prefix.

Algorithm.

1. Run the original untargeted attack for T iterations.
2. Lock onto the leading non-true class

$$t = \arg \max_{k \neq y} P(k \mid x'_T).$$
3. Continue the same attack, but maximize the target confidence $P(t \mid x')$.
4. Stop when the classifier prediction is no longer y .

- Irreversible lock: no repeated exploration.
- Stable for $T \in \{5, \dots, 500\}$: rankings appear early.

3. Experimental Protocol

Models	AlexNet, ResNet-18, VGG-16, ResNet-50, ViT-B/16
Attacks	SimBA, Square Attack with CE loss, Bandits
Modes	Untargeted baseline, targeted oracle, OTS
Budget	$\epsilon = 8/255$ in L_∞ , 10K iterations
Scale	100 ImageNet images per model, 4,500 standard-model runs
Metrics	Success rate and censored mean iterations

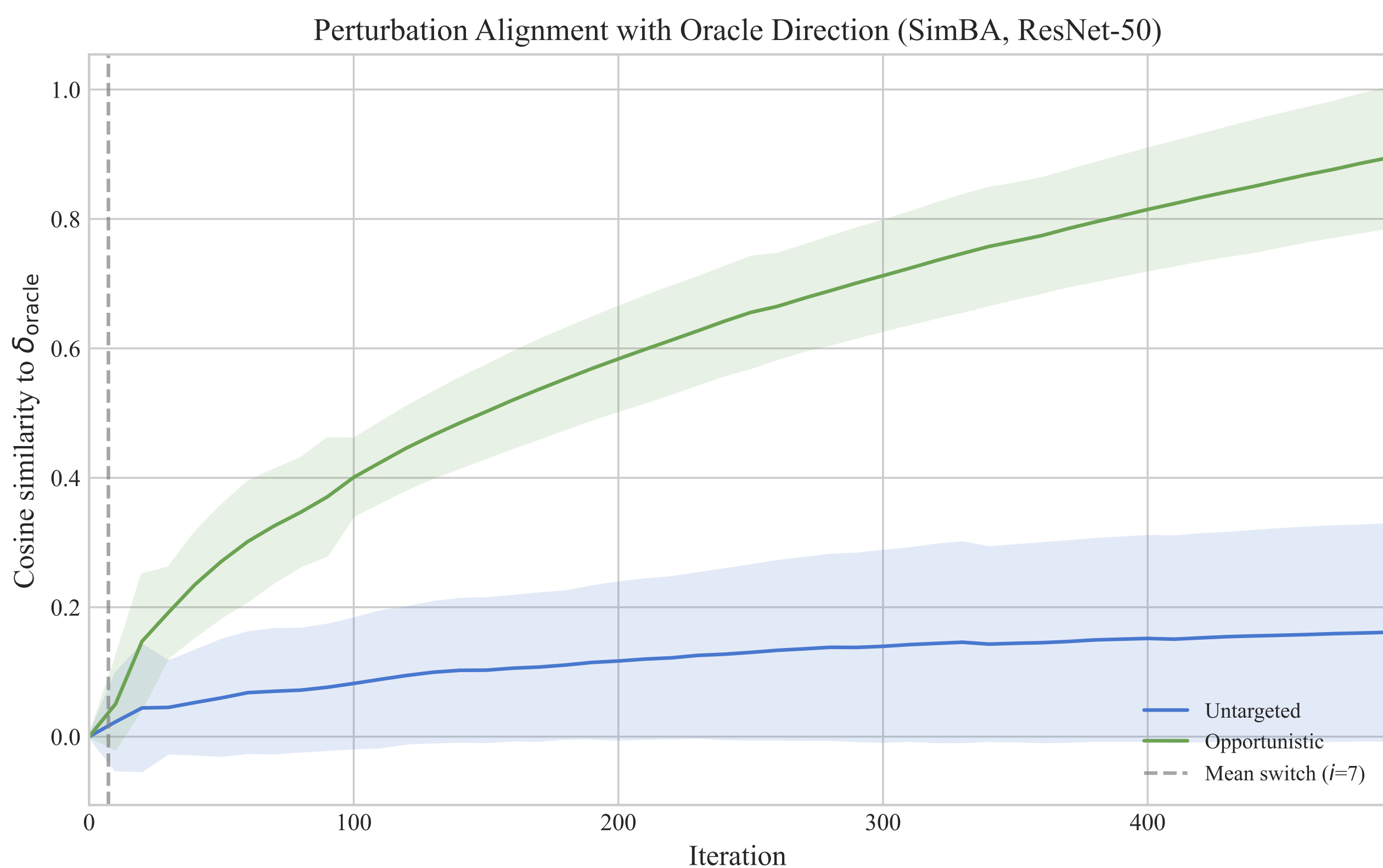
Oracle target: same-seed untargeted endpoint, used as a trajectory-specific ceiling.

4. When Should OTS Help?

Attack	Direction source	Expected OTS effect
SimBA	No target tracking	Large gain from early commitment
Square CE	y -only CE	Margin-like gain
Bandits	Gradient estimate	Small or negative gain

- Helps when the base loss drifts.
- Can hurt when the base attack already has a direction.

5. Mechanism



- OTS aligns with the oracle direction: $0.174 \rightarrow 0.865$ cosine.
- Fixed t mimics the margin competitor:

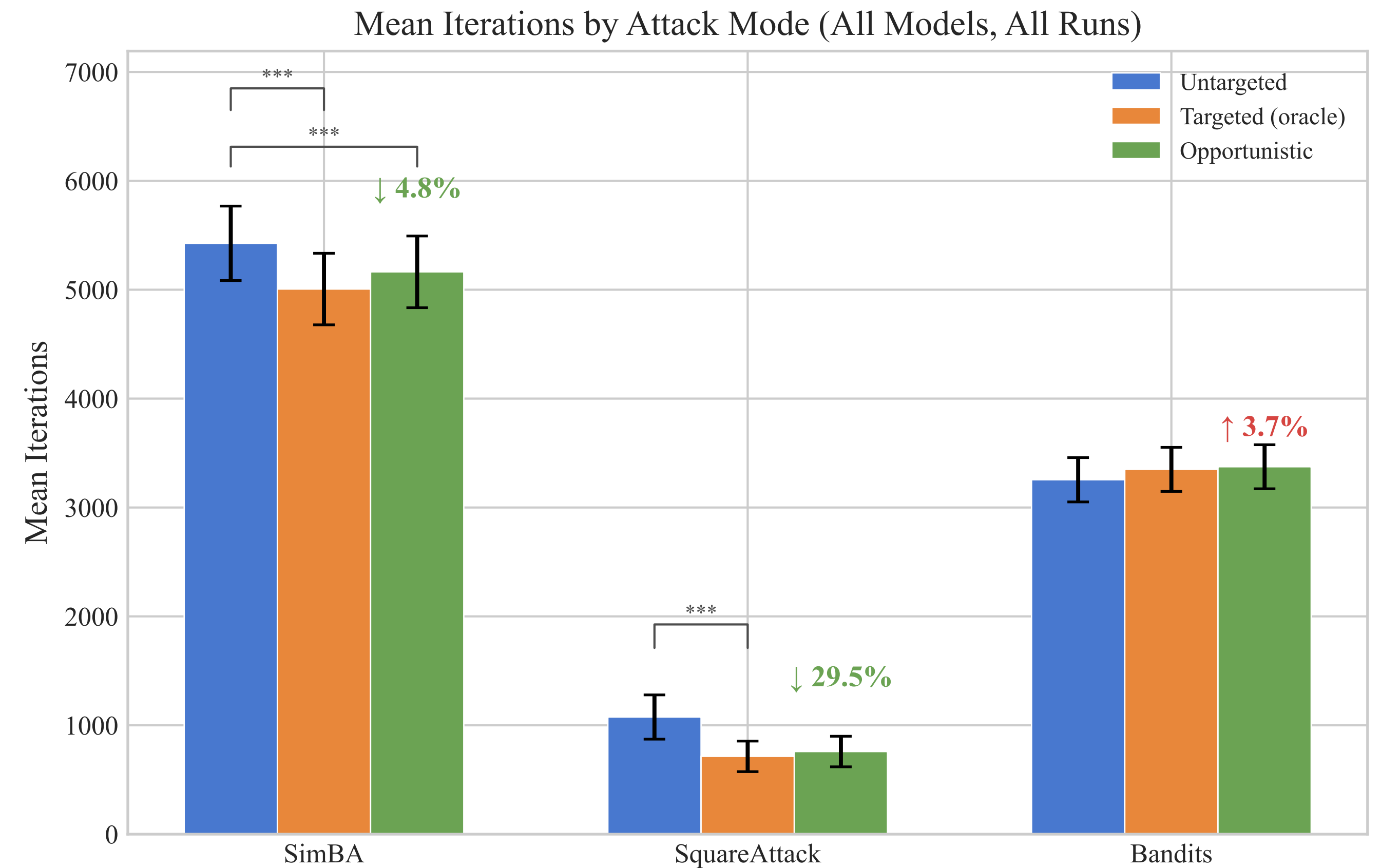
$$\mathcal{L}_{\text{margin}} = f_y(x') - \max_{k \neq y} f_k(x').$$

6. Target-Rule Ablation

Target rule	SimBA	Square CE
Random non-true class	14%	53%
Clean-image argmax	79%	96%
Short exploration	80%	99%

- Random targets are catastrophic.
- SimBA: clean ranking works; Square CE: short exploration helps.

7. Main Results



Attack	Untargeted	OTS	Oracle
SimBA	67%	75%	76%
Square CE	96%	99%	99%
Bandits	37%	34%	35%

+27 pp SimBA success on ResNet-50 **43%** relative iteration reduction **0** gradient access

Main finding.

- Near-oracle behavior for SimBA and Square CE.
- Bandits gets worse: it already follows an estimated gradient.

8. Lock-In Dynamics



- Untargeted: suppresses the true class, then drifts.
- OTS: locks class 988 and keeps increasing it.
- Only a stable competitive target is needed.

9. Loss Interpretation

Square Attack, ResNet-50	Success	Mean iter.
CE untargeted	85.0%	2,601
CE + OTS	98.0%	1,791
CE oracle	99.0%	1,865
Margin untargeted	98.7%	1,453
Margin + OTS	98.7%	1,583

- CE drift is repaired by OTS.
- Margin loss already tracks the competitor, so OTS is redundant.

10. Limits & Future Work

- Robust ImageNet models: flat success across variants.
- No medium-difficulty regime where commitment pays off.

Future work.

- CIFAR-10, L_2 budgets, randomized defenses.
- Adaptive re-locking if the target drops in rank.

11. Takeaways

Take-away.

- Explore briefly.
- Lock the strongest non-true class.
- Useful only when the base attack drifts.